

FUNCHAL, Portugal

WMO: 085220

Lat: 32.6475N

Lon: 16.8925W

Elev: 184

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	54.9	56.0	34.8	29.8	66.7	38.5	34.6	62.3	15.9	64.9	14.0	65.0	3.8	20	0.213

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	81.8	69.2	80.2	69.3	79.0	68.9	72.1	78.9	71.2	77.9	70.4	77.1	5.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	110.5	76.3	68.8	106.5	76.0	67.8	102.7	75.7	35.9	79.1	35.1	78.0	34.4	77.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.3	10.4	8.7	DB	52.5	88.3	1.8	4.3	51.1	91.4	50.1	93.9	49.0	96.3	47.7	99.5
			WB	45.3	74.0	1.3	0.9	44.3	74.6	43.6	75.1	42.9	75.5	42.0	76.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.9	62.9	62.2	63.2	64.1	66.2	69.6	72.6	74.6	74.1	71.9	67.7	64.7	(6)
	DBStd	4.98	2.42	2.50	2.68	2.48	2.51	2.03	1.98	2.04	1.90	2.51	2.90	2.28	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	320	72	84	69	42	13	0	0	0	0	0	8	32	(9)
	CDD50	6517	400	342	408	424	503	588	701	763	722	677	532	457	(10)
	CDD65	1363	7	6	12	16	52	138	236	298	272	213	90	23	(11)
	CDH74	3169	1	4	27	36	42	99	518	1093	859	415	64	11	(12)
	CDH80	212	0	0	0	9	11	3	29	90	45	23	2	0	(13)
(14) Wind	WSAvg	3.6	4.1	4.1	4.3	3.7	3.3	2.8	2.7	2.7	3.1	3.6	4.1	4.3	(14)
(15) Precipitation	PrecAvg	24.3	3.5	3.0	2.5	1.5	0.8	0.4	0.1	0.1	0.9	3.5	3.9	3.8	(15)
	PrecMax	59.4	11.8	18.9	9.9	8.2	4.3	2.7	1.3	0.7	7.6	9.7	12.0	11.7	(16)
	PrecMin	14.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)
	PrecStd	8.3	3.1	3.1	2.1	1.6	0.9	0.6	0.2	0.2	1.5	2.6	2.7	3.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	73.7	77.5	79.6	78.3	78.7	83.3	87.3	83.7	82.8	79.4	75.3	(19)
		MCWB	59.8	56.8	56.1	59.5	62.6	68.9	69.4	66.1	69.9	64.6	61.4	57.1	(20)
	2%	DB	70.0	70.1	72.2	72.2	73.9	76.4	79.7	81.7	81.0	79.3	75.6	71.7	(21)
		MCWB	60.4	58.9	59.0	61.3	64.8	67.5	69.5	71.1	69.9	68.4	65.1	61.8	(22)
	5%	DB	68.5	68.3	69.5	69.9	72.3	74.9	78.0	80.1	79.6	77.7	73.8	70.0	(23)
		MCWB	59.7	58.8	60.1	60.8	63.9	66.6	68.7	70.4	69.4	67.9	64.6	61.4	(24)
	10%	DB	67.1	66.8	67.9	68.4	70.7	73.6	76.6	78.8	78.3	76.2	72.3	68.7	(25)
		MCWB	59.0	58.0	59.4	59.9	62.8	65.7	67.9	69.6	68.7	67.1	63.8	60.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.7	63.9	64.3	64.7	67.4	70.2	71.9	73.6	73.5	72.5	70.0	66.2	(27)
		MCDB	68.8	67.5	69.0	71.0	73.5	76.7	79.3	80.7	79.8	77.8	74.0	69.6	(28)
	2%	WB	63.2	62.0	62.8	63.3	65.9	68.8	70.7	72.3	72.0	70.9	68.3	64.8	(29)
		MCDB	67.7	66.9	67.7	69.5	72.1	75.2	77.8	79.6	78.3	76.3	73.0	68.9	(30)
	5%	WB	61.8	60.6	61.7	62.2	64.8	67.5	69.8	71.3	71.0	69.8	66.6	63.5	(31)
		MCDB	66.9	66.2	67.0	68.3	70.7	73.7	76.5	78.6	77.5	75.5	71.9	68.3	(32)
	10%	WB	60.4	59.4	60.5	61.1	63.7	66.5	68.9	70.4	69.9	68.6	65.1	62.1	(33)
		MCDB	65.9	65.4	66.4	67.3	69.7	72.7	75.6	77.6	76.7	74.6	70.9	67.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.0	9.4	9.1	8.8	8.3	8.6	9.3	9.4	9.1	8.6	8.1	(35)
		MCDBR	9.6	10.7	11.3	10.8	10.5	9.7	10.2	10.8	10.7	10.5	9.8	9.4	(36)
	5% WB	MCWBR	7.6	8.0	7.9	7.6	7.2	6.5	6.7	6.6	7.0	7.3	7.1	7.3	(37)
		MCWBR	8.9	9.4	9.7	10.0	9.6	9.1	9.4	10.0	9.9	9.2	8.9	8.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.341	0.362	0.389	0.378	0.381	0.380	0.389	0.393	0.390	0.375	0.366	0.347	(40)
		taud	2.552	2.473	2.388	2.436	2.451	2.477	2.455	2.466	2.491	2.526	2.511	2.546	(41)
	Ebn at Noon		279	283	282	289	287	286	283	281	278	275	267	270	(42)
		Edh at Noon	26	31	36	36	36	35	36	35	33	29	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		906	1166	1564	1921	2106	2249	2246	2096	1761	1339	963	812	(44)
	RadStd		45	78	79	69	130	107	93	71	57	58	56	40	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page